UNIVERSITY OF DHAKA

Department of Electrical and Electronic Engineering

FabLab DU — Project Documentation

Design, Fabrication, and Flight Testing of a Low-Cost ESP32-Based Quadcopter UAV

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1. Abstract

This project documents the design, fabrication, and flight testing of a low-cost quadcopter unmanned aerial vehicle (UAV) built at FabLab DU, University of Dhaka. The primary objectives were to develop hands-on understanding of open-source flight controller firmware (ESP-FC/Betaflight) and to produce a functional, budget-constrained drone platform using an ESP32 microcontroller as the flight controller. The resulting system achieved stable hover, controlled manoeuvring, and a practical flight time of approximately 12–13 minutes, at a total build cost of approximately BDT 7,500 (approx. USD 61). The platform is intentionally designed for future extensibility with additional sensors, camera modules, and payloads.

2. Introduction and Objectives

Commercial flight controllers based on STM32 microcontrollers are the industry standard for hobby and research UAVs, but their cost can be prohibitive for student projects in resource-constrained settings. This project explored whether a commodity ESP32 microcontroller, paired with the open-source ESP-FC firmware and the widely used Betaflight Configurator interface, could serve as a viable, low-cost flight controller for a functional quadcopter.

The work was carried out at FabLab DU under the supervision of Dr. Md. Shafiul Alam, with a target budget of BDT 8,200 (approx. USD 66). The specific objectives were:

- To understand and implement a Betaflight-compatible flight controller using ESP32 and ESP-FC firmware.
- To fabricate and assemble a complete quadcopter (F450 Quad-X frame) within the stated budget.
- To configure PID-based stabilisation via Betaflight Configurator and validate stable flight.
- To produce a documented, upgradeable platform suitable for future integration of sensors, cameras, and autonomous modules.

3. System Architecture and Hardware

3.1 Configuration

The drone follows a standard Quad-X configuration with four BLDC motors at the corners of an F450 frame. Pilot commands from a FlySky FS-i6 transmitter are received by the FS-iA6B receiver, which sends PPM/SBUS data to the ESP32. The ESP32 reads IMU data from the MPU6050 via I2C, computes PID corrections, and outputs PWM signals (1000–2000 μs) to four 30A ESCs, which in turn drive the motors.

3.2 Component Specifications

Component	Specification
Flight Controller	ESP32 DevKit V1 (240 MHz dual-core)
IMU	MPU6050 (I2C, gyroscope + accelerometer)
Frame	F450 Quad-X
Motors	A2212 1000KV BLDC
ESCs	30A (PWM protocol)
Propellers	8045 (8 × 4.5 inch)
Battery	3S 11.1V 2200mAh LiPo (XT60 connector)
Transmitter	FlySky FS-i6
Receiver	FlySky FS-iA6B
Firmware	ESP-FC (Betaflight Configurator v10.10)

3.3 ESP32 GPIO Mapping

Function	GPIO Pin
Motor 1	GPIO 13
Motor 2	GPIO 12
Motor 3	GPIO 14
Motor 4	GPIO 27
SDA (MPU6050)	GPIO 21
SCL (MPU6050)	GPIO 22
Receiver Input (PPM)	GPIO 35
Battery Monitor (ADC)	GPIO 36

A 30kΩ/10kΩ resistor voltage divider scales the 3S LiPo voltage (up to 12.6 V) down to a maximum of 3.15 V, within the ESP32 ADC safe range.

4. Engineering Design

4.1 Weight and Thrust

The total all-up weight (AUW) of the assembled drone is 870 g (0.87 kg). With a target thrust-to-weight ratio (TWR) of 2:1, the system achieved an actual TWR of 2.8:1, corresponding to a maximum total thrust of approximately 23.93 N (~2,440 gf). This confirms adequate lift margin for stable hover and basic manoeuvring.

4.2 RPM and Hover Characteristics

Using the motor KV rating and full-charge battery voltage (12.6 V):

Maximum RPM = KV × V = 1000 × 12.6 = 12,600 RPM

The hover RPM is approximately 7,524, corresponding to a hover throttle of 55–60%.

4.3 Estimated Flight Time

Based on battery capacity and average current draw, estimated flight times are approximately 16 minutes for sustained hover and 12–13 minutes for mixed flight. Observed free-flight duration was approximately 13 minutes, consistent with estimates.

4.4 Electrical Wiring

The power architecture follows a Battery → PDB → ESCs → Motors layout. Wire gauges were selected per current load: 14 AWG for battery mains, 16 AWG for ESC-to-motor lines, and 26 AWG for signal wiring.

5. Firmware Configuration and PID Tuning

5.1 Firmware

ESP-FC firmware was flashed to the ESP32 via ESP Web Flasher over USB. The firmware is fully compatible with Betaflight Configurator v10.10, enabling GUI-based configuration. Key capabilities used include a PID loop at up to 2 kHz, PWM ESC protocol, PPM receiver support, ACRO and ANGLE flight modes, and hardware failsafe. Motor and sensor GPIO assignments were applied via Betaflight CLI resource mapping commands.

5.2 PID Control Structure

The flight controller uses a standard cascaded PID structure: an outer loop for angle control and an inner loop for angular rate control. The PID output is computed as:

$$\mathbf{u(t) = K_{\square} \cdot e(t) \; + \; K_i \cdot \int e(t)dt \; + \; K_x \cdot [de(t)/dt]}$$

Final PID values after bench and tethered tuning:

Axis	P	I	D
Roll	42	45	18
Pitch	46	50	22
Yaw	35	45	0

6. Assembly and Testing

6.1 Assembly Procedure

Assembly followed the sequence: F450 frame construction → motor mounting → ESC installation → power distribution wiring → mounting of ESP32 and MPU6050 → signal wiring → propeller installation (last, for safety). Pre-flight checks included motor direction verification, gyroscope calibration, RC range testing (~50 m), failsafe validation, centre-of-gravity balance, and battery state verification.

6.2 Issues Encountered and Resolutions

Issue	Resolution
Hover oscillations	Reduced proportional (P) gain
Yaw drift	IMU alignment correction via Betaflight
ESC desync	Increased motor idle throttle
High-throttle vibrations	Propeller balancing

7. Results

Parameter	Result
Stable Hover	Achieved
Maximum Altitude	~15 m
Observed Flight Time	~13 min
Hover Stability	±5°
Landing	Controlled
Thrust-to-Weight Ratio	2.8:1
Total Build Cost	~BDT 8,200 (approx. USD 66)

8. Future Improvements

The current platform intentionally provides a minimal, functional baseline and is well-suited for incremental expansion. Planned and suggested improvements include integration of a GPS module (e.g., Neo-6M) for position hold and return-to-home capability; addition of a barometric altimeter (e.g., BMP280) for altitude hold mode; mounting a lightweight FPV camera with video transmitter for first-person view flight; incorporating ultrasonic or LiDAR sensors for obstacle detection and low-altitude hold; upgrading ESCs to support DSHOT protocol for faster, more noise-resistant motor communication; and exploring autonomous waypoint navigation using the ESP32's onboard Wi-Fi for telemetry and mission planning. The ESP32's remaining GPIO capacity and computational headroom make it capable of supporting several of these additions without a hardware change to the flight controller itself.

9. Conclusion

This project demonstrated that a fully functional, Betaflight-compatible quadcopter flight controller can be constructed using a commodity ESP32 microcontroller and the open-source ESP-FC firmware, at a fraction of the cost of commercial alternatives. The assembled drone achieved stable hover, controlled flight to approximately 15 m, and a flight duration of ~13 minutes, with all primary objectives met. The hands-on process of configuring Betaflight, mapping GPIO resources via CLI, and tuning PID parameters provided significant practical learning in embedded flight control systems. The documented platform and component choices provide a reproducible, upgradeable baseline for future UAV development activities at FabLab DU.

References

- ESP-FC Firmware Repository — GitHub (open-source, ESP32 Betaflight-compatible firmware)
- Betaflight Configurator v10.10 — Official Betaflight Project
- InvenSense MPU-6050 Product Specification, Rev 3.4
- FlySky FS-i6 / FS-iA6B User Manual
- Espressif Systems — ESP32 Technical Reference Manual